

Sugam Panthi

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Core Competencies

AI/ML: LLM Applications, Conversational Memory Systems, RAG Pipelines, VLMs, Predictive Modeling, Fine-tuning
Architecture & Infrastructure: AWS (ECS, Fargate, Lambda, ECR), Docker, Terraform, CI/CD, Microservices
Frameworks & Tools: Go, Python, LangChain, PyTorch, Pinecone, Flask, React, C/C++

Industry Experience

Co-Founder & CTO, *MagnoliaEd LLC*

Apr 2025 – Present

- Built and shipped an AI-powered education platform with personalized instructor chatbots and role-based dashboards; deployed across multiple college classrooms.
- Architected a hybrid monolithic system on AWS Fargate using Docker and Terraform, scaling chat, analytics, and auth independently across 1,000+ concurrent sessions.
- Established CI/CD pipelines with GitHub Actions, cutting deployment friction and automating test coverage across the full stack.
- Secured pre-seed funding through Co-Builders powered by Microsoft; led investor outreach and placed in multiple pitch competitions.

AI & ML Fellow, *Prediction 3D*

May 2025 – August 2025

- Developed predictive models in PyTorch on multimodal construction-site data, combining images and sensor readings to generate safety and productivity insights for field teams.
- Integrated AI features into client-facing applications with LangChain and Pinecone, collaborating within a 20-engineer team to deploy ML capabilities into production workflows.
- Contributed to a microservice-based data pipeline separating ingestion, preprocessing, and model inference, improving maintainability and deployment flexibility.

AI Research Assistant, *University of Southern Mississippi*

Nov 2024 – Present

- First-authored two papers: showed that fixed RAG compression hides reader upgrades and reverses model rankings across 20 readers, and that scoring-target choice alone flips LLM memory benchmark conclusions on fixed retrieval output.
- Designed and ran evaluation pipelines across four QA benchmarks and one summarization dataset, auditing compression–reader interaction over ten domain-method settings.
- Built reproducible analyses across LoCoMo, LongMemEval, Mem0, and MemoryOS; separately modeled U.S. plastic waste trends with LSTM networks, contributing to two journal paper submissions.

Engagements & Recognition

- **Research:** Fixed RAG Compression Collapses Measured Reader Scaling. arXiv preprint (June 2026) - arXiv:2606.21807
- **Research:** Same Ranking, Different Winner: How Scoring Targets Shape LLM Memory Benchmarks. arXiv preprint (May 2026) - arXiv:2605.24060
- **Writing:** Technical Blog Webpage
- **Pitch Competition Winner**, Mississippi Polymer Institute May 2025
- **Hackathon Winner**, Davidson College January 2025
- **Hackathon Winner**, University of Southern Mississippi November 2024

Projects

Pali (Link) — *Open-Source Memory Infrastructure for LLMs*

- Local-first, inspectable memory runtime for LLM agents; exposes REST, MCP, and dashboard interfaces over a shared memory core written in Go.
- Multi-tenant storage and retrieval with lexical + vector search, candidate fusion, and configurable embedding/scoring providers.
- Pluggable backends (SQLite, Qdrant, Neo4j); benchmarked across LoCoMo, LongMem, and BEAM.

deepseek-go (Link) — *Open-Source Go SDK*

- Most popular Go client for DeepSeek-compatible APIs; 325 stars, 38 forks, 45+ issues closed, 30+ PRs.
- Shipped structured releases through v1.3.3 covering streaming support, reliability fixes, and contributor onboarding.
- Maintained issue triage and community contributions; used in production by multiple companies.

Cohesion — **AI-Powered SQL Assistant** (Link) — *Hackathon Winner*

- Built an AI-driven schema generation tool with a custom speed-optimized CSV format; implemented backend services using Go, Gin, and Docker.

Education

The University of Southern Mississippi, Honors College
B.S. in Computer Science

Hattiesburg, MS
GPA: 4.0 / 4.0